

Market Risk Analysis of Tesla (TSLA): Volatility Forecasting, VaR, ES, and Option Risk

Section 1. Introduction

This report evaluates the downside risk of an investment in Tesla (TSLA), a stock characterised by high volatility and non-normal return behaviour. The analysis focuses on 1-day 99% Value at Risk (VaR) and Expected Shortfall (ES), key measures used in market-risk regulation and internal capital modelling (Basel Committee, 2013). VaR identifies the loss threshold exceeded on 1 out of every 100 trading days, while ES captures the average loss when this threshold is breached, providing a deeper assessment of tail severity.

The objective is to compare several prominent approaches to measuring short-horizon risk and to examine how their assumptions shape the resulting estimates. The methods include parametric techniques, semi-parametric procedures that incorporate time-varying volatility, and a tail-focused Peaks Over Threshold (POT) model from Extreme Value Theory. A full re-pricing Monte Carlo simulation is also applied to a 1-month at-the-money call option to capture non-linear risk.

The report is organised as follows. Section 2 outlines the methodologies used to compute VaR and ES. Section 3 describes the data and key distributional properties of Tesla returns. Section 4 presents the empirical findings for the stock and the option. Section 5 concludes by summarising the main insights, highlighting limitations, and suggesting areas where model robustness could be improved.

Section 2. Method

This report quantifies the 1-day 99% Value at Risk (VaR) and Expected Shortfall (ES) for Tesla stock and a call option using distinct methodologies for both unconditional and conditional volatility. Using multiple approaches mitigates model risk and reveals sensitivity to assumptions (Jorion, 2007). All VaR and ES are computed using daily simple returns and losses are defined as $L_t = -r_t$ so that large negative returns correspond to large positive losses.

2.1 Volatility Estimation

Volatility enters every model as the scale parameter of return risk. Two measures are employed.

Unconditional Volatility (σ_{uncond}) is the sample standard deviation of daily simple returns over the full five-year period, assuming a constant variance process.

$$\sigma_{uncond} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (r_i - \bar{r})^2}$$

Note: Where r_i is return on the day i , $\bar{r}$ is the mean return, and n is the number of observations.

Conditional Volatility (σ_{cond}) is estimated using the EWMA(0.94) model, which updates volatility according to recent return shocks and captures volatility clustering.

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

Note: Where σ_t^2 is the conditional variance for day t , r_{t-1} is the return on the previous day, and λ is a decay factor, specified as 0.94, following the RiskMetrics framework (J.P. Morgan, 1996).

2.2 Variance-Covariance (VCV) Approach

The VCV approach assumes normally distributed returns with zero mean, so VaR and ES depend solely on the chosen volatility input.

The 1-day 99% Value-at-Risk (VaR) and Expected Shortfall (ES) are defined as:

$$VaR_{0.99} = \sigma \Phi^{-1}(1 - 0.01)$$

$$ES_{0.99} = \sigma \frac{\phi(\Phi^{-1}(1 - 0.01))}{0.01}$$

Note: Where σ is the daily volatility, $\Phi^{-1}(p)$ is the inverse of the standard normal cumulative distribution function (the Z-score), and $\phi(\cdot)$ is the standard normal probability density function.

2.3 Standardised Historical Simulation (Unconditional and Standardised)

We adopt the Historical Simulation approach of computing risk measures directly from the empirical distribution of past simple returns while adjusting for time-varying volatility. Returns are standardised using:

$$z_t = \frac{r_t - \mu}{\sigma_t}$$

Note: Where r_t is the daily simple return, μ is the sample mean, and σ_t is the volatility. This implementation does not use a rolling window. Therefore the first 250 volatility estimates are excluded to avoid distortion from model initialisation. The 1% quantile of the standardised returns is then re-scaled to obtain VaR. ES is computed as the mean of standardised returns below this quantile.

This approach preserves the empirical loss distribution while adapting to current volatility conditions.

2.4 Monte Carlo Simulation (Geometric Brownian Motion)

The Monte Carlo simulation approach uses a large number of randomly generated scenarios to model potential future price changes. 200,000 1-day price paths are simulated under the assumption that the stock follows a Geometric Brownian Motion (GBM) process:

$$S_{t+1} = S_t \times \exp [(r_f - 0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z]$$

Note: Where S_t is the current stock price, r_f is the annualised risk-free rate (4.364%), σ is the daily volatility, $\Delta t = 1/252$ represents one trading day, and $Z \sim N(0,1)$ is a random shock.

The daily volatility input is the EWMA conditional volatility forecast for 5 August 2025. This choice reflects the most recent state of market risk and responds to volatility clustering observed in Tesla's returns, making it more appropriate for a 1-day ahead risk forecast than the unconditional volatility. Implied volatility is not used, as it reflects option market expectations over a 1-month horizon and is employed instead in the Black-Scholes re-pricing of the call option in Section 2.6.

2.5 Peaks Over Threshold (POT)

The POT model uses Extreme Value Theory to estimate tail risk by modelling excess losses above a high threshold u , chosen as the 95th percentile of empirical losses. Excesses are fitted to a Generalised Pareto Distribution with shape parameter ξ and scale parameter β . The 1-day 99% VaR and ES are:

$$VaR_{0.99} = u + \frac{\beta}{\xi} \left[\left(\frac{0.01}{\frac{n_u}{n}} \right)^{-\xi} - 1 \right]$$

$$ES_{0.99} = \frac{VaR_{0.99} + (\beta - \xi u)}{1 - \xi}$$

Note: Where $\frac{n_u}{n}$ is the probability of exceeding the threshold.

2.6 Option VaR and ES (Full Re-pricing)

For the at-the-money call option expiring on 5 September 2025, risk is measured using full re-pricing. Each simulated stock price from the GBM approach is re-valued using the Black-Scholes model:

$$C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$$

Note: Where $d_1 = \frac{\ln(S/K) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$, S is spot price and K is strike price. The Black-Scholes pricing uses the implied volatility of 46% obtained from the CBOE option chain on 4 August 2025, and the time to maturity T is measured in years from the valuation date to the option's expiry.

1-day option returns are computed from the re-priced values, and VaR and ES are calculated from the simulated return distribution. This captures convexity and avoids reliance on linear approximations.

Section 3. Data

The analysis is based on historical daily data for TSLA stock, sourced from Yahoo Finance. The sample period covers 5 years of daily trading data. The analysis uses daily frequency data and the primary data series used is the daily adjusted closing price. Figure 1 illustrates the price evolution, which characterises significant trends and periods of high volatility.

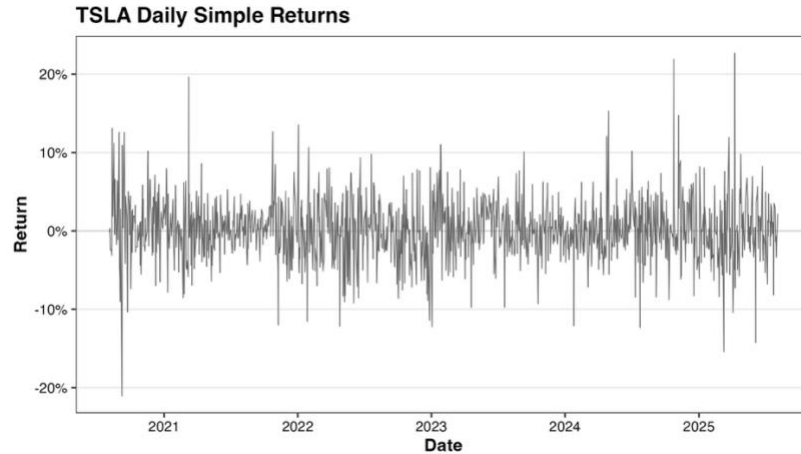
Figure 1: TSLA Adjusted Close Price (3rd August 2020 - 4th August 2025)



Note: The figure shows an upward trend with several periods of volatility, particularly in late 2022, and early 2025.

Figure 2 describes the daily price series converted into a series of daily simple returns ($\frac{P_t}{P_{t-1}} - 1$). We observe major volatility clustering in late 2022 and early 2025 coincide with broader technology-sector corrections and shifts in interest rate expectations, consistent with macroeconomic drivers of Tesla's market risk. The alternating periods of high and low volatility motivate the use of time-varying volatility models.

Figure 2: Daily Simple Returns for TSLA



Note: The return series exhibits volatility clustering, meaning recent volatility influences future risk (Engle, 1982).

Table 1 presents the key descriptive statistics for the daily simple returns of TSLA over the observation period. The standard deviation of daily returns is 4%. This figure serves as the unconditional volatility for the investment, representing the average daily risk over the 5 year period. The Mean daily return is approximately 0.17%. This aligns with market behaviour for daily data and supports the assumption of zero mean for the VaR and ES calculations. The Kurtosis of 6.15 is the most critical statistic. A normal distribution has a kurtosis of 3. A value significantly greater than 3 indicates a leptokurtic distribution. The presence of fat tails in returns is well-documented in empirical finance, with extreme movements occurring far more frequently than the normal model predicts (Danielsson and de Vries, 1997). It strongly suggests that risk models assuming a normal distribution (such as the Variance-Covariance and Monte Carlo approaches) may underestimate the likelihood of large losses.

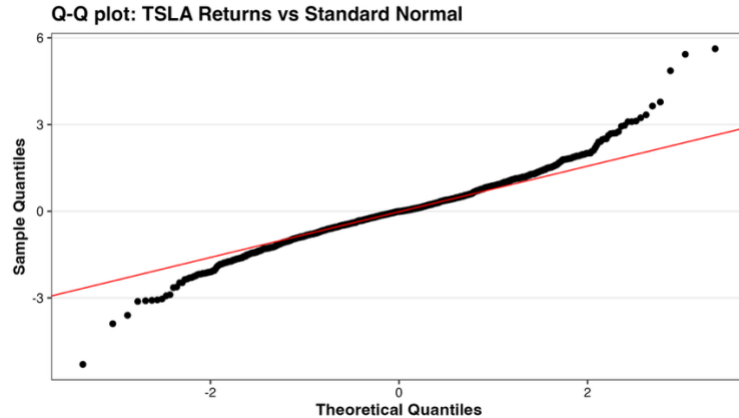
Table 1: Descriptive Statistics of Daily Simple Returns

n	Mean Daily Return	Sample Standard Deviation	Skewness	Kurtosis	Minimum Daily Return	Maximum Daily Return
1256	0.17%	4%	0.2697	6.1518	-21.06%	22.69%

Note: The raw dataset originally contained 1,258 rows, including a zero-price placeholder for 5th August 2025 (used for out-of-sample forecasting) and the first trading day with no prior return observation.

The fat-tailed nature of the returns is confirmed by the Quantile-Quantile (QQ) plot in Figure 3. This plot compares the quantiles of the empirical return data against the theoretical quantiles of a normal distribution. The heavy-tailed nature of the returns indicates that extreme losses occur more frequently than a normal model would imply. Consequently, risk estimates based on parametric methods that assume normality are likely to underestimate downside risk. This motivates the use of non-parametric and Extreme Value Theory approaches.

Figure 3: Q-Q Plot of TSLA Returns vs. Standard Normal Distribution



Note: the data points deviate significantly from the straight diagonal line, particularly at the lower and upper tails. This shape is a visual signature of a fat-tailed distribution, corroborating the high kurtosis value in Table 1.

Finally, the analysis incorporates ancillary data as of 4th August 2025: a 1-month Treasury Bill rate of 4.364% and option pricing data from the CBOE. For the option risk analysis, the call option with a strike price of \$310 was selected, as it is the closest available strike to the TSLA closing price of \$309.26 on that day.

Section 4. Results

This section presents and interprets the results of the Value-at-Risk (VaR) and Expected Shortfall (ES) methodologies applied to Tesla (TSLA) stock and a 1-month at-the-money call option. The analysis compares how different modelling assumptions (normality, time-varying volatility, and tail behaviour) affect the estimation of market risk.

4.1 Risk of the Underlying Stock (TSLA)

The 1-day 99% VaR and ES results for Tesla stock are summarised in Table 2.

Table 2. Tesla Stock 99% 1-Day VaR and ES Estimates

Method	1-day 99% VaR	1-day 99% ES
VCV uncond	9.32%	10.67%
VCV cond	8.03%	9.20%
HS uncond	11.68%	14.62%
HS cond	10.07%	12.60%
MC (GBM)	7.77% (lowest)	8.87% (lowest)
POT (GPD)	9.983%	12.75%

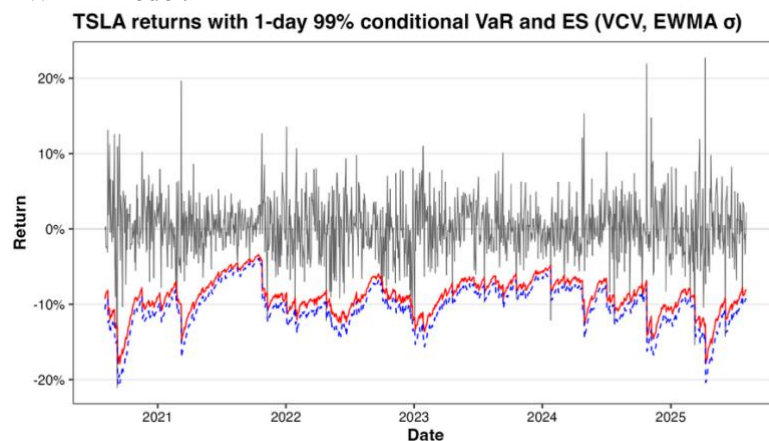
Note: All figures report 1-day 99% VaR and ES in simple return terms, expressed as positive magnitudes of loss relative to the TSLA price on 4 August 2025. VCV uncond and cond use unconditional volatility (4%) and conditional volatility forecast (3.45%) respectively. HS uncond and cond are Historical Simulation estimates with fixed and time-varying volatility. MC (GBM) is a Monte Carlo simulation assuming a GBM process with σ_{cond} . POT (GPD) applies to Peaks over Threshold Extreme Value model with 95th percentile loss threshold. A VaR of 10% indicates a 1% probability of losing more than 10% value in a single trading day.

We first examine the two volatility inputs that underpin all models. The unconditional volatility is 4%. This measure reflects the long-run average variability of Tesla's price and is unaffected by recent market movements. On the other hand, the conditional volatility forecast from the EWMA(0.94) model for 5 August 2025 is 3.45%. This is lower than the unconditional figure because the most recent observations fall within a period of relative stability for Tesla. The conditional estimate therefore provides a more accurate input for next-day risk, since it responds to volatility clustering and places greater weight on recent information.

The results reveal that estimated risk levels vary depending on the chosen model and its assumptions. Under the VCV framework, the unconditional VaR and ES is 9.32% and 10.67% respectively, compared to 11.68% and 14.62% for the unconditional HS approach. The higher HS values are consistent with the high kurtosis (6.15) observed in the return distribution.

The same pattern persists in the conditional models, which are more relevant for next-day forecasting. The conditional VCV VaR of 8.03% and ES of 9.2% is lower than the conditional HS VaR of 10.07% and ES of 12.6%, even though both use the same volatility forecast. This pattern shows that heavy tails dominate the differences across models rather than volatility level alone. Figure 4 illustrates how the 99% conditional VaR and ES evolves through time when the EWMA volatility forecast is used.

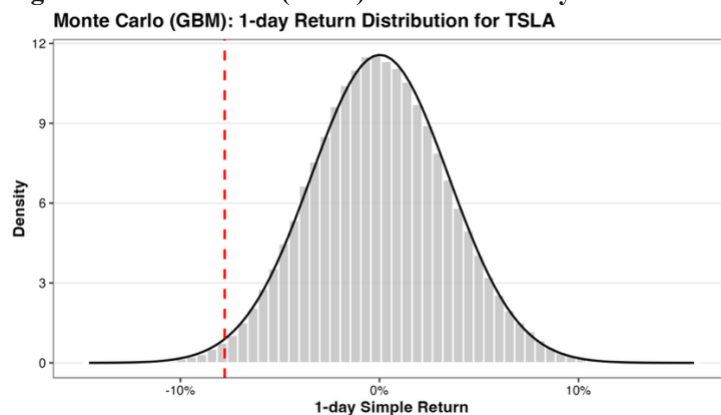
Figure 4. Tesla Daily Returns with Time-Varying 1-Day 99% Conditional VaR Derived from the EWMA Model.



Note: The red line shows the evolving VaR band and the blue line shows Expected Shortfall (ES). We examine widening during volatile periods and narrowing during stable market conditions.

The Monte Carlo and Peaks Over Threshold (POT) models provide further insight into the tail behaviour of Tesla's return distribution. The Monte Carlo approach, based on a Geometric Brownian Motion (GBM) process, produces the lowest VaR estimate of 7.77% and ES estimate of 8.87%. Figure 5 displays the simulated 1-day return distribution from this model. The histogram's shape directly results from the GBM assumption that daily shocks are normally distributed. While this model captures the continuous-time nature of price changes, it still inherits the same thin-tail limitation as the VCV approach and consequently underestimates the likelihood of large losses.

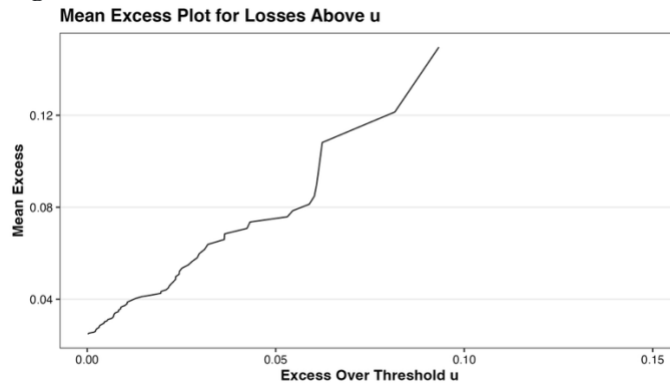
Figure 5. Monte Carlo (GBM) Simulated 1-Day Return Distribution for TSLA



Note: The histogram follows a Normal shape, and the red line marks the 99% VaR threshold. This explains why both VaR and ES under the Monte Carlo GBM model are the lowest across all methods: the thin-tailed normal shocks artificially compress the probability of extreme losses.

In contrast, the Peaks Over Threshold (POT) model, grounded in Extreme Value Theory (EVT), explicitly focuses on the tail of the loss distribution. Using the 95th percentile of losses as the threshold, the model fits a Generalised Pareto Distribution (GPD) to the excesses. The resulting 1-day 99% VaR and ES values are 9.98% and 12.75% respectively. The fitted Generalised Pareto parameters are $\xi = 0.076$ and $\beta = 0.0226$. The positive ξ confirms heavy tails, since $\xi > 0$ implies Pareto-type tail decay, meaning extreme Tesla losses occur more frequently than under thin-tailed models such as the normal distribution. However, because only 5% of observations lie above the threshold, the effective tail sample is small, so ξ and β are estimated with sampling uncertainty, making POT results sensitive to threshold choice. Figure 6 below presents the mean excess plot used to validate the selection of the 95th percentile threshold.

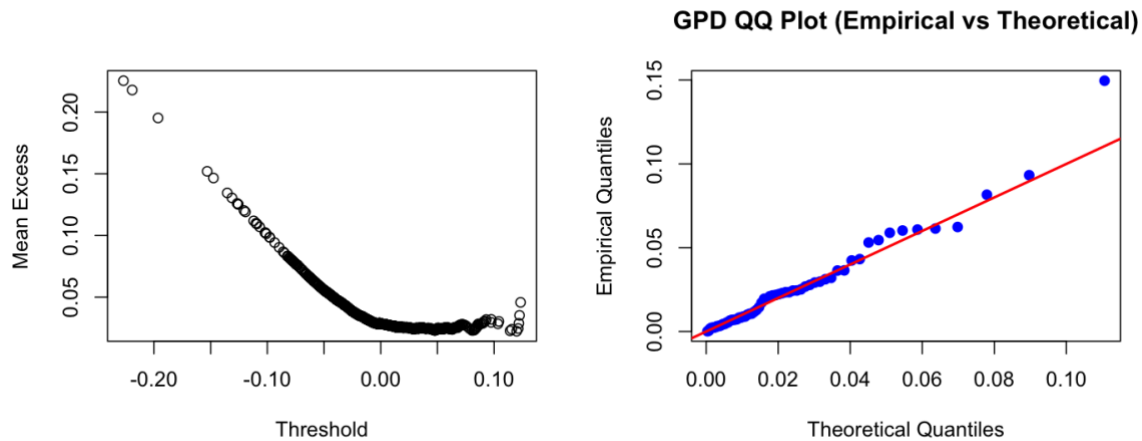
Figure 6. Mean Excess Plot for Losses Above the Threshold



Note: The mean excess plot displays the average size of losses conditional on exceeding a range of candidate thresholds. For a Generalised Pareto Distribution, the mean excess function is approximately linear above an appropriate threshold. The upward-sloping region beyond $u \approx 0.05$ indicates heavy-tailed behaviour and supports the choice of the 95th percentile threshold for the POT estimation.

With the threshold validated, diagnostic plots confirm that the fitted GPD adequately captures the tail behaviour. Figure 7 summarises the full set of POT diagnostic checks.

Figure 7. Diagnostic Plots for the Peaks Over Threshold (GPD) Model



Note: (Left) Mean-excess plot showing a linear trend above the 95th percentile threshold. (Right) QQ plot confirming the GPD provides a good fit to the extreme loss data.

The sensitivity analysis was performed and findings are summarised in Table 3. It reports the effects of changing both the volatility input used in the Monte Carlo simulation and the decay factor applied in the EWMA model for conditional volatility. The finding reveals a significant divergence between historical and market-implied risk. The Monte Carlo VaR based on the backward-looking EWMA volatility was 7.77%. However, when using the forward-looking implied volatility of 46% derived from option prices, the VaR fell to 6.56%. This occurs because the implied annual volatility of 46% corresponds to a daily volatility of approximately 2.9%, which is lower than the EWMA-based daily volatility of 3.45%.

Table 3. Sensitivity Analysis

Component	Specification	σ daily	1-day 99% VaR	1-day 99% ES
Monte Carlo (GBM)	Conditional EWMA volatility	3.45%	7.77%	8.87%
Monte Carlo (GBM)	Implied volatility 46%	2.9%	6.56%	7.46%
VCV (EWMA λ sensitivity)	$\lambda = 0.90$	-	7.44%	8.52%
VCV (EWMA λ sensitivity)	$\lambda = 0.94$ (baseline)	-	8.03%	9.2%
VCV (EWMA λ sensitivity)	$\lambda = 0.97$	-	9.21%	10.56%

Note: Replacing conditional EWMA volatility with market implied volatility of 46% corresponds to a daily volatility of 2.9%, lowering the corresponding estimates.

For the conditional VCV model, varying the EWMA decay parameter λ from 0.90 to 0.97 produces 1-day 99% VaR estimates between 7.44% and 9.21%, with ES ranging from 8.52% to 10.56%. Higher λ values lead to slower volatility decay, producing correspondingly higher VaR and ES estimates.

4.2 Risk of the Call Option

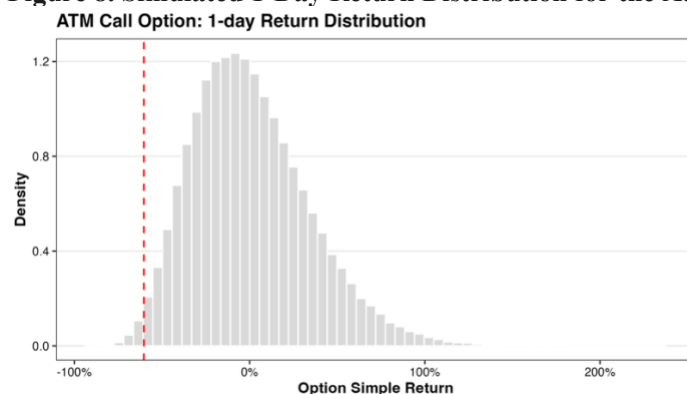
The 1-day 99% VaR and ES for the at-the-money (ATM) call option with a strike price of \$310 were estimated using a full re-pricing Monte Carlo simulation. Each of 200,000 simulated stock prices generated under the GBM process, was re-valued using the Black-Scholes model with an implied volatility of 46%. The results are reported in Table 4.

Table 4. 1-Day 99% VaR and ES for the TSLA At-The-Money Call Option

TSLA Price	Strike (K)	Option Mid-Price	Implied Volatility (Annual)	1-day 99% VaR	1-day 99% ES
309.26	310	17.13	46%	60.35%	65.63%

Note: VaR and ES are expressed as simple return losses relative to the option's mid-price on 4 August 2025. The magnitudes reflect the high leverage, convexity and short maturity of the near-ATM call.

The VaR of 60.35% implies that, with 99% confidence, the option's 1-day loss will not exceed approximately 60% of its market value. This magnitude far exceeds that of the underlying stock, whose VaR ranges between 8 and 12%, illustrating the substantial leverage embedded in option positions. Because options provide non-linear payoffs, small changes in the underlying asset can cause disproportionately large percentage changes in option value. This convexity amplifies both gains and losses and explains why option VaR figures are several times larger. Figure 8 illustrates the simulated 1-day return distribution for the option. The distribution is highly asymmetric and positively skewed. This skewed distribution contrasts sharply with the symmetric shape of the underlying stock's return distribution and underscores the need for full re-pricing to accurately capture non-linear risk.

Figure 8. Simulated 1-Day Return Distribution for the At-The-Money TSLA Call Option (K = 310)

Note: The left tail extends steeply to around -100%, corresponding to the total loss of the option premium, while the right tail tapers slowly, representing the theoretically unlimited profit potential of a call option. The red vertical line indicates the 99% VaR threshold located in the left tail.

Section 5. Conclusion

This report presents the 1-day 99% Value at Risk (VaR) and Expected Shortfall (ES) for Tesla stock and a 1-month at-the-money call option using a range of parametric, semi parametric and tail focused methods. Across all specifications, the results show that model choice materially affects the magnitude of estimated downside risk. Parametric models produced lower estimates than HS and POT, consistent with the heavy-tailed nature of Tesla's returns.

For the underlying stock, the most credible 1-day 99% VaR estimates cluster around 10%, with corresponding ES values in 12~13%. Conditional models based on the EWMA volatility forecast are generally lower than their unconditional counterparts, which is consistent with the relatively calm volatility regime observed near the end of the sample. The Standardised Historical Simulation VaR provides a sensible benchmark for day to day risk monitoring, since it combines a realistic empirical loss distribution with an up to date measure of volatility. For capital planning and assessment of more extreme scenarios, the POT based ES is a more conservative and theoretically robust indicator of tail risk because it focuses directly on the behaviour of the worst losses. However, EVT estimates remain sensitive to threshold choice and the small effective sample size in the tail, which can introduce sampling uncertainty into POT based VaR and ES.

The call option analysis confirms that derivative positions embody substantially greater risk than the underlying stock. This finding supports the standard risk management practice of treating option exposures separately from cash equity positions and using full repricing approaches rather than linear approximations.

The sensitivity analysis highlights that VaR and ES are not fixed constants but are conditional on modelling choices. Reasonable changes in these assumptions move the estimated VaR for Tesla stock by several tens of basis points. This underlines the importance of model governance, including routine sensitivity checks and periodic recalibration when volatility regimes shift.

Several limitations follow from the methodology. The analysis is based on a single asset, a fixed 5 year historical window and a 1-day risk horizon, and it assumes that past return dynamics and the fitted GPD tail continue to hold in the future. The GBM specification also restricts shocks to be normally distributed, which is known to be restrictive for equities like Tesla. Future work could extend the framework to multi asset portfolios, incorporate GARCH or stochastic volatility models in place of GBM, and backtest the competing VaR and ES models. Subject to these caveats, the report provides a characterisation of short horizon downside risk for Tesla and illustrates clearly how different modelling assumptions feed into risk estimates that would be used in practice for capital allocation and stress testing.

Section 6. References

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